

Galois Correspondence

Arham Lodha

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Problem 1. Recall our Ext homework, where we showed (notation from the homework, I hope) that $\text{Ext}_K^L(g)$ has size at most $|L/K|$ with equality iff for each element a of L , $g(p_a)$ splits completely, with distinct roots (p_a the minimal poly). Look back at the argument and see this equivalent to an apparently weaker condition: $L = K[a_1, \dots, a_s]$ such that $g(p_{a_i})$ splits as above – so if this holds for a set of generators, iff it holds for all a in L .

Proof. Suppose $L = K[a_1, \dots, a_s]$ where $g(p_{a_i})$ splits completely and has distinct roots in A . Define, $L_n = K[a_1, \dots, a_n] = L_{n-1}[a_n]$.

Base Case: $L_1 = K[a_1]$. Since $g(p_{a_1})$ splits as above, $|\text{Ext}_K^{L_1}(g)| = [L_1 : K] \implies \forall a \in L_1, g(p_a)$ splits, by field Symmetries homework. . .

Inductive Hypothesis: Assume that $\forall n < k$, that the theorem holds for L_n

Inductive Step: Let $n = k$. $L_k = L_{k-1}[a_k]$. $\forall \tilde{g} \in \text{Ext}_K^{L_{k-1}}(g)$. By the Field Symmetries Homework, since $g(p_{a_n})$ splits completely, $\tilde{g}(\tilde{p}_{a_n})$ splits completely, since $\tilde{p}_{a_n} | p_{a_n}$, thus $|\text{Ext}_{L_{k-1}}^{L_k}(\tilde{g})| = [L_{k-1} : L_k]$. Thus the map, $\text{Ext}_K^{L_k}(g) \rightarrow \text{Ext}_{L_{k-1}}^{L_k}(g)$ is surjective and $|\text{Ext}_K^{L_k}(g)| \leq [L_{k-1} : L_k][L_{k-1} : K] = [L_k : K]$. However since, $|\text{Ext}_{L_{k-1}}^{L_k}(\tilde{g})| = [L_{k-1} : L_k]$ and by the inductive Hypothesis and field Symmetries homework, $[L_{k-1} : L_k] = |\text{Ext}_{L_{k-1}}^{L_k}(\tilde{g})|$. Thus the inequality is actually a equality and $|\text{Ext}_K^{L_k}(g)| = [L_k : K]$. Thus by field Symmetries homework, $\forall l \in L_n, g(p_l)$ splits.

Thus theorem holds for L .

$\Leftarrow$: If it holds for all a it holds for generators. □

Problem 2. Say $L = K[a]$, show Gal is equal to the set of roots of $p_a \in L$. In particular, it has order at most

$\deg(L/K)$ with equality iff the min poly has distinct roots.

Proof. The canonical map between $\text{Gal}(L/K)$ and the roots of $p(X)$ is the following $h \rightarrow h(a)$.

Suppose $\forall h, k \in \text{Gal}(L/K)$, $h(a) = k(a) \implies \forall x + ya, h(x + ya) = x + yh(a) = x + yk(a) = k(x + ya) \implies h = k$.

$\forall r$ in the roots of $g(p)[X]$, since $K[r] \cong K[x]/(p_r) \cong K[x]/(p_a) \cong K[a]$, thus there is an automorphism in Gal which sends $a \rightarrow r$.

Thus the map is bijection.

Thus $|\text{Gal}(L/K)|$ is at equal the number of unique roots of p in L which is at most the number of unique roots of p which is at most $\deg p = [L : K]$. Thus $|\text{Gal}(L/K)| \leq [L : K]$.

Suppose $|\text{Gal}(L/K)| = [L : K]$. Then the number of unique roots of p in L is $[L : K]$. $\deg(p) = [L : K]$. Thus the number of unique roots of p in L is equal to the number of possible roots of p (or the degree). Thus every root is unique and p has no double root. Thus since the number of roots of p in L is equal to the number of roots of p , all roots are in L . Thus p has all roots in L and no double roots.

Suppose p has all roots in L and no double roots. The number of unique roots of p in L is equal to $\deg p = [L : K]$. Since $|Gal(L/K)|$ equals the number of number of unique roots of p in F , $|Gal(L/K)| = [L : K]$. $\square$

Problem 3. We say L/K is Galois if G has order $\deg(L/K)$. show L/K is Galois iff it is separable and a splitting field (meaning its generated by the roots of a polynomial over K).

Proof. $\implies$: Suppose L/K is Galois. Then $Gal(L/K) = [L : K]$, thus $\forall a \in L$, the minimal polynomial, p_a , splits and is separable in L . Thus L is separable and splitting.

$\impliedby$: Suppose L is separable and splitting over K . By field Symmetries, $|Gal(L/K)| = [L : K]$. Thus L/K is Galois. $\square$

Problem 4. Suppose H is a subgroup of G , and $L^H = K$. Show that $\forall a \in L$, its minimal poly has roots exactly the orbit Ha . Show L/K is primitive, and $H = G$.

Proof. Suppose $H \leq G$. $\forall a \in L$, let $g(x) := \prod_{\alpha \in H \cdot a} (x - \alpha)$. g has distinct roots because its roots are by definition elements of the set Ha and thus distinct. Furthermore since all elements of H are automorphisms, $H \cdot a \subseteq L$, and thus g splits completely over L . $\forall h \in H$, $h(g(x))$ just reorders the product, thus $h(g(x)) = g(x)$. Thus $g(x)$ is H invariant. Thus the coefficients of g are H invariant and thus live L^H . Suppose, $g(x) = h(x)k(x)$ where $h(x), k(x) \in K[X]$. Since a is a root of $g(x)$, a must be a root of $h(x)$ or $g(x)$. WLOG, suppose a is a root of $h(x)$. By homomorphism properties, $\forall \sigma \in H$, $h(\sigma(a)) = \sigma(h(a)) = 0$, thus $\sigma(a)$ is a root of $h(x)$. Thus all roots of $g(x)$ are roots of $h(x)$. Thus $k(x)$ must be unit and $h(x) = g(x)$. Thus $g(x)$ is a irreducible over L^H . Thus g is the min polynomial of a . Thus L/L^H is separable and splitting. By primitive element theorems, $\exists \alpha \in \overline{K} \ni L = K[\alpha]$. Since L/L^H is separable and is a splitting field, $|Gal(L/K)| = [L : K] = \deg p_\alpha \leq |H|$. However $H \leq Gal(L/K) \implies |H| \leq |Gal(L/K)| \implies |Gal(L/K)| = |H| \implies H = Gal(L/K)$. $\square$

Problem 5. Show L/L^H is always Galois (with no assumptions on L/K).

Proof. Suppose $H \leq G$. $\forall a \in L$, let $g(x) := \prod_{\alpha \in H \cdot a} (x - \alpha)$. g has distinct roots because its roots are by definition elements of the set Ha and thus distinct. Furthermore since all elements of H are automorphisms, $H \cdot a \subseteq L$, and thus g splits completely over L . $\forall h \in H$, $h(g(x))$ just reorders the product, thus $h(g(x)) = g(x)$. Thus $g(x)$ is H invariant. Thus the coefficients of g are H invariant and thus live L^H . Suppose, $g(x) = h(x)k(x)$ where $h(x), k(x) \in L^H[X]$. Since a is a root of $g(x)$, a must be a root of $h(x)$ or $g(x)$. WLOG, suppose a is a root of $h(x)$. By homomorphism properties, $\forall \sigma \in H$, $h(\sigma(a)) = \sigma(h(a)) = 0$, thus $\sigma(a)$ is a root of $h(x)$. Thus all roots of $g(x)$ are roots of $h(x)$. Thus $k(x)$ must be unit and $h(x) = g(x)$. Thus $g(x)$ is a irreducible over L^H . Thus g is the min polynomial of a . Thus L/L^H is separable and splitting. Thus L/L^H is Galois. By primitive element theorems, $\exists \alpha \in \overline{L^H} \ni L = L^H[\alpha]$. $\forall u, v \in H \ni u\alpha = v\alpha \implies uv^{-1}\alpha = \alpha$. $\forall c, d \in L^H$, $uv^{-1}(c + d\alpha) = c + d\alpha \implies uv^{-1} = e$. Thus $u = v$. Thus $\deg p_a = |H \cdot a| = |H|$. Thus, $[L : K] = |H|$.

$\forall h \in H$, h by definition fixes elements of L^H and is a automorphism of L . Thus $h \in G$. Thus $H \leq Gal(L/L^H)$. But since $|H| = |G| \implies H = Gal(L/L^H)$.

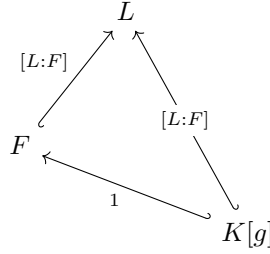
Thus the map from subgroups to subfields is injective. $\square$

Problem 5b. Now the other direction, we assume L/K is Galois. explain why $L = K[a]$ for some a . Take intermediate field F . explain why $L = F[a]$.

Proof. Since L/K is galois its seperable. Thus by primitive elements theorems, $\exists \alpha \in \overline{K} \ni L = K[\alpha]$.

Let $K \subseteq F \subseteq L$. $L = K[\alpha] \subseteq F[\alpha]$. $\forall f \in F[\alpha]$, $f = \sum_{n=0} f_n \alpha^n$ where $f_i \in F$. Since $F \subseteq K[\alpha] \implies f_i = \sum_{m=0} k_{m,i} \alpha^m$ for $k_{k,i} \in K$. Thus $f = \sum_{n=0} \sum_{m=0} k_{m,i} \alpha^{m+n}$. Thus $f \in K[\alpha]$. Thus $F[\alpha] \subseteq K[\alpha]$. Thus $F[\alpha] = K[\alpha]$.

$g_1, \dots, g_k \in F$ and $K \subseteq F$, thus $K[g_1, \dots, g_k] = K[g] \subseteq F$. Thus g must divide the minimum polynomial of α over $K[g]$, but $g \in K[g][X]$, so g is the minimum polynomial of α over $K[g]$. Thus $[L : F] = [L : K[g]]$.



Thus $[F : K[g]] = 1$. Thus $F = K[g]$. □

Problem 5c. Explain why G^F the subgroup of G that preserves g (ie takes the poly g to itself, equivalently, fixes all the coefficients). Explain why this is the same as those which permute the roots of g . Explain why an element of G is completely determined by where it sends " α ". Explain why G^F acts transitively on the roots of g . Explain why $|G^F| = \text{degree}(g)$. Now explain why L/L^{G^F} has degree $\text{deg}(g)$. Conclude $L^{G^F} = L$.

Proof. $G^F = \text{Gal}(L/F)$, thus since $F = K[g]$ and G^F fixes F that means it fixes everything in F which means it fixes the coefficients of G which means it fixes g .

Thus $\forall x \in L$, $\sigma(g(x)) = g(\sigma(x))$. $\forall \sigma \in G^F$, let r be a root of g , $0 = \sigma(g(r)) = g(\sigma(r)) \implies \sigma(r)$ is a root of g . Thus σ permutes the roots. $\forall \sigma \in \text{Gal}(L/K)$ which permutes the roots of g . Let the roots of g be $g_1, \dots, g_n$. Thus $g(x) = a(x - g_1) \dots (x - g_n)$. Thus $\sigma(g(x)) = a(x - \sigma(g_1)) \dots (x - \sigma(g_n))$ since σ permutes the roots, $\sigma(g(x)) = a(x - \sigma(g_1)) \dots (x - \sigma(g_n)) = a(x - g_1) \dots (x - g_n) = g(x)$. Thus it preserves g and $\sigma \in G^F$.

$\forall \sigma \in G$, since $L = K[\alpha]$. $\forall x \in L$, $x = c + d\alpha$ for $c, d \in K$. $\sigma(x) = c + d\sigma(\alpha)$, since σ fixes K , and the only choice is where to send α . Thus σ is determined by where to send α .

Since L/L^{G^F} is galois. For any roots a and b . The homomorphism which sends a to b and fixes $K[g]$, is a valid automorphism of G and thus is also in G^F . Thus G^F acts transitively.

Since G^F acts transitively on the roots of g and faithfully on the roots of g and is determined where it sends a , thus $|G^F| = \text{deg } g$, or the number of roots of g .

g is irreducible, because its roots are the orbit of G^F . Thus g is the minimal polynomial over L^{G^F} . Thus $[L : L^{G^F}] = \text{deg } g$. Thus $[F : L^{G^F}] = 1$. Thus $L^{G^F} = F$. □

Problem 6. Note G acts naturally on subsets of L , and this induces an action on the intermediate fields. These are in bijection with subgroups, Show the corresponding action is by subgroups.

Proof. G acts on the subsets of L and the subgroups of G .

$$\forall S \in P(L), \forall g \in G, g \cdot S := \{g(s) | s \in S\}$$

$$\forall H \leq G, \forall g \in G, g \cdot H = gHg^{-1}.$$

WTS: $\forall g \in G, L^{g \cdot H} = L^{gHg^{-1}} = g \cdot L^H$.
 $\forall g \in G$,
 $\forall l \in g \cdot L^H, l = g(m)$ for some $m \in L^H, \forall h \in H, ghg^{-1}l = (ghg^{-1})(g(m)) = gh(m) = g(m) = l$.
Thus $l \in L^{gHg^{-1}}$. Thus $g \cdot L^H \subseteq L^{gHg^{-1}}$.
 $\forall k \in L^{gHg^{-1}}, \forall h \in H$, Thus $k = ghg^{-1}k \implies g^{-1}k = hg^{-1}k$. That means, $g^{-1}k \in L^H$. Thus
 $gg^{-1}k = k \in g \cdot L^H$.
Thus $g \cdot L^H \subseteq L^{gHg^{-1}} \implies g \cdot L^H = L^{gHg^{-1}}$ $\square$

Problem 6b. Conclude that L^H is preserved by G iff $H \trianglelefteq G$. What is the quotient G/H ?

Proof. $\implies$: Suppose L^H is preserved by G , then $\forall g \in G, L^H = g \cdot L^H = L^{gHg^{-1}}$. By injectivity of Galois Correspondence, $H = gHg^{-1}$. Thus $H \trianglelefteq G$.

$\impliedby$: Suppose, $H \trianglelefteq G$. Thus $\forall g \in G, g \cdot L^H = L^{gHg^{-1}} = L^H$.

Let $\varphi : G \rightarrow \text{Gal}(L^H/K)$. By field symmetries homework, $\ker \varphi = \text{Gal}(L/L^H) = H$. $G/H \cong \text{Gal}(L^H/K)$. $\square$

Problem 7. Take F an arbitrary subfield. Let $H := G^F$. Take any $g : F \rightarrow L$ which is the identity on K . Explain how H acts on $\text{Ext}_F^L(g)$. Explain why this action is "simply transitive" – meaning transitive, with no fixed points. $\text{Ext}_F^L(g)$ is not usually a group but its a "torsor" for a group – that just means there is a group that acts on it simply transitively – so picking an element of it puts it in bijection with the group.

Proof. Define the action of H as follows: $\forall h \in H, \forall p \in \text{Ext}_F^L(g), h \cdot p = h \circ p$.

$\forall p \in \text{Ext}_F^L(g), p$ is a field map and thus injective and thus surjective and thus a automorphism of F .

$\forall p, q \in \text{Ext}_F^L(g), h : L \rightarrow L$, where $l \rightarrow q(p^{-1}(l))$. Since $q, p \in \text{Aut}(F) \implies q \circ p^{-1} \in \text{Aut}(F)$. Furthermore since q, p^{-1} fix elements of $K, q \circ p^{-1} \in \text{Gal}(L/K)$. $\forall f \in F, h(f) = q(p^{-1}(f)) = g(g^{-1}(f)) = f$. thus $h \in H$. Furthermore, $h \circ p = q$. Thus H acts transitively.

Since L galois wrt to K and thus $L = K[\alpha] = F[\alpha]$, every map in G is uniquely determined by where it sends α . Thus $\forall h \in H$ and $\forall p \in \text{Ext}_F^L(g)$ where $(h \circ p) = p$. Thus, $h(p(p^{-1}(\alpha))) = p(p^{-1}(\alpha)) = \alpha$ thus $h(\alpha) = \alpha$ thus since everything G is uniquely determined by where its sends $\alpha, h = \text{id}$. Thus there are no fixed points. $\square$

Problem 8. Suppose we have a Galois extension as above, and two subgroups H_1 and H_2 of G . What is $L^{H_1 \cap H_2}$?

Proof. $H := \text{Gal}(L/L^{H_1}L^{H_2})$.

$\forall \alpha \in L^{H_1}L^{H_2}, \alpha = \sum_{i=1}^n a_{h_1,i} b_{h_2,i}$ where $a_{h_1,i} \in L^{h_1}$ and $b_{h_2,i} \in L^{h_2}, \forall h \in H_1 \cap H_2$, since $h(\alpha) = h(\sum_{i=1}^n a_{h_1,i} b_{h_2,i}) = \sum_{i=1}^n h(a_{h_1,i}) h(b_{h_2,i}) = \sum_{i=1}^n a_{h_1,i} b_{h_2,i} = \alpha$. Thus $h \in H$. $H_1 \cap H_2 \subseteq H$

Since $L^{H_1} \subseteq L^{H_1}L^{H_2}, L^{H_2} \subseteq L^{H_1}L^{H_2}, \forall h \in H, h$ fixes L^{H_1} and L^{H_2} . Since $H_1 = \text{Gal}(L/L^{H_1})$ and $H_2 = \text{Gal}(L/L^{H_2})$, and h is a automorphism of L which fixes L^{H_1} and L^{H_2} thus, $h \in H_1$ and $h \in H_2$. Thus $h \in H_1 \cap H_2$. Thus $H \subseteq H_1 \cap H_2$.

By Galois Correspondence $H \cong H_1 \cap H_2 \iff L^{H_1}L^{H_2} = L^{H_1 \cap H_2}$. $\square$

Problem 8b. How about $L^{\langle H_1, H_2 \rangle}$ (meaning the subgroup generated by both H_1 and H_2)?

Proof. $H = \text{Gal}(L/L^{H_1} \cap L^{H_2})$

$\forall h \in \langle H_1, H_2 \rangle$. $\forall \alpha \in L^{H_1} \cap L^{H_2}$, by definition α is an element fixed by both H_1 and H_2 . Since $h = h_{1,1}h_{2,1} \dots h_{1,n}h_{2,n}$, $h(\alpha) = h_{1,1}h_{2,1} \dots h_{1,n}h_{2,n}(\alpha) = \alpha$. Thus $h \in H$ and $\alpha \in L^{\langle H_1, H_2 \rangle}$. $L^{H_1} \cap L^{H_2} \subseteq L^{\langle H_1, H_2 \rangle}$.

$\forall \alpha \in L^{\langle H_1, H_2 \rangle}$, since $H_1, H_2 \leq \langle H_1, H_2 \rangle$. α fixed by H_1 and H_2 thus $\alpha \in L^{H_1} \cap L^{H_2}$. Thus $L^{\langle H_1, H_2 \rangle} \subseteq L^{H_1} \cap L^{H_2}$.

Thus, $L^{\langle H_1, H_2 \rangle} = L^{H_1} \cap L^{H_2}$ □

Problem 9. Lets say we have a Field extension B over K (B stands for big), and inside we have two fields L_1 and L_2 , whose intersection is K , both Galois over K . Let L_1L_2 be the smallest subfield of B that contains both. Show L_1L_2 is Galois over K , and that its Galois group is naturally the product of the $\text{Gal}(L_i/K)$ (or if this is wrong, explain why).

Proof. Since L_1 and L_2 are galois, they are seperable and thus primitive by primitive element theorem. $L_1 = K[a]$ and $L_2 = K[b]$. Thus $L_1L_2 = K[a, b]$. Since L_1 is galois, the minimum polynomial of a is seperable and has roots in L_1 and thus in L_1L_2 . Similarly since L_2 is galois, the minimum polynomial of b is seperable and has roots in L_2 and thus in L_1L_2 . By Problem 1, thus since, a, b are seperable and splitting, $K[a, b]$ is seperable and splits. Thus $L_1L_2 = K[a, b]$ is galois.

$\varphi : \text{Gal}(L_1L_2/K) \rightarrow \text{Gal}(L_1/K) \times \text{Gal}(L_2/K)$ where $\psi \rightarrow (\psi|_{L_1}, \psi|_{L_2})$. $\forall a \in \ker \varphi$, $\varphi(a) = (id, id) \implies a|_{L_1} = id$ and $a|_{L_2} = id$ thus $a = id$ and φ is injective. Since $L_1 \cap L_2 = K$, a and b are independent. Thus, $\forall g \in \text{Gal}(L_1L_2/K)$, g is completely determined by which roots to send a and b to. Thus $|\text{Gal}(L_1L_2/K)| = [L_1 : K][L_2 : K]$. $\text{Gal}(L_1/K) \times \text{Gal}(L_2/K) = |\text{Gal}(L_1/K)| |\text{Gal}(L_2/K)| = [L_1 : K][L_2 : K] = |\text{Gal}(L_1L_2/K)|$. Thus, $\text{Gal}(L_1L_2/K) \cong \text{Gal}(L_1/K) \times \text{Gal}(L_2/K)$ □

Problem 10. 10) Say L/K is finite Galois, and we have subfields M and N with $M \cap N = K$, and $MN = L$ (MN means the smallest subfield containing both M and N). Finally, suppose M over K is also Galois. Express $\text{Gal}(L/K)$ as a semi-direct product. Show $\text{Gal}(L/N)$ is naturally iso to $\text{Gal}(M/K)$ (or if this is wrong, explain why)

Proof. Let $H = \text{Gal}(L/M)$ and $J = \text{Gal}(L/N)$. $M \cap N = L^H \cap L^J = L^{\langle H, J \rangle} = K$. $L^H L^J = L^{H \cap J} = L \implies \text{Gal}(L/L) = H \cap J = id$. $\text{Gal}(L/K) = \langle H, J \rangle$. Since M/K is galois, there is a homomorphism $\text{Gal}(L/K) \rightarrow \text{Gal}(M/K)$, (restriction map), and the kernel is H , by problem 6. Thus H is normal. Thus, $\text{Gal}(L/K) = H \rtimes J$. J is the kernel of $H \rtimes J \rightarrow (H \rtimes J)/H$, Thus $J \cong \text{Gal}(L/K)/\text{Gal}(L/M) \cong \text{Gal}(M/K)$, by the restriction above. □